



Mini Project Report - 17

**Bachelor of Computer Application
Semester – IV**

Sub: Full Stack Development

Topic: CRUD Operation in MDB

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February 2026

Introduction to MongoDB (MDB):

This project titled “**CRUD Operations in MongoDB (MDB)**” explains the basic database operations performed using MongoDB Community Server and MongoDB Compass.

MongoDB is a NoSQL, document-oriented database developed by **Dwight Merriman**, along with co-founders Eliot Horowitz and Kevin P. Ryan. It was designed to handle large amounts of unstructured data and provide high performance, scalability, and flexibility.

MongoDB stores data in JSON-like format called BSON (Binary JSON), which makes it different from traditional relational databases that use tables and rows.

In this project, we study MongoDB and perform CRUD operations (Create, Read, Update, Delete) using both MongoDB shell commands and MongoDB Compass GUI.

The objective of this project is to understand how data is stored, retrieved, updated, and deleted efficiently using MongoDB.

Requirements of Community and Compass:

MongoDB Community Server is the core database engine that stores and manages data. Below are the minimum and recommended requirements.

Operating System Requirements

MongoDB Community Server supports:

- Windows 10 or later (64-bit)
- macOS 10.14 or later
- Linux distributions (Ubuntu, Debian, Red Hat, CentOS)

Note: MongoDB only supports 64-bit operating systems.

Hardware Requirements

Minimum Requirements:

- 4 GB RAM
- 2 GHz Dual-Core Processor
- At least 1 GB free disk space

Recommended Requirements (for better performance):

- 8 GB RAM or more
- Multi-Core Processor
- SSD storage for faster read/write operations

Software Requirements

- 64-bit Operating System
- Administrator privileges for installation
- Internet connection (for downloading packages)
- MongoDB Service should be enabled during installation

Requirements for MongoDB Compass

MongoDB Compass is the graphical user interface (GUI) tool used to interact with the MongoDB database visually.

Operating System Requirements

MongoDB Compass supports:

- Windows 10 or later (64-bit)
- macOS 10.14 or later
- Linux (Ubuntu, Debian)

Hardware Requirements

Minimum:

- 4 GB RAM
- 500 MB free disk space

Recommended:

- 8 GB RAM for handling large datasets smoothly

Software Requirements

- MongoDB Community Server must be installed first
- Internet connection for download
- Compatible 64-bit system

5.4 Why These Requirements Are Important

Meeting these requirements ensures:

- Smooth installation
- Proper database performance
- Faster query execution
- No connection errors
- Stable CRUD operations

If system requirements are not met, MongoDB may run slowly or fail to start properly.

Difference and Uses Between Community and Compass

MongoDB Community Server and MongoDB Compass are two different components of the MongoDB ecosystem, and both serve different purposes.

MongoDB Community Server is the core database engine. It is responsible for storing, managing, and processing data. It runs as a background service on the system and listens for database requests on the default port 27017. All CRUD operations, indexing, data storage, and query processing are handled by the Community Server. Without it, no data can be stored or retrieved because it is the main database system.

On the other hand, MongoDB Compass is a graphical user interface (GUI) tool that allows users to interact with the MongoDB database visually. It does not store data itself. Instead, it connects to the MongoDB Community Server and provides an easy way to manage databases, collections, and documents.

Another major difference is the method of interaction. MongoDB Community Server is typically accessed using command-line tools such as MongoDB Shell. This requires knowledge of query syntax and commands. In contrast, MongoDB Compass provides buttons, menus, and forms to perform operations like inserting, updating, and deleting documents. It is more user-friendly, especially for beginners.

MongoDB Community Server is mandatory for running MongoDB because it acts as the actual database engine. MongoDB Compass is optional but highly recommended because it simplifies database management.

In simple words, the Community Server stores and processes data, while Compass helps users view and manage that data visually.

Uses of MongoDB Community Server

MongoDB Community Server is used for the following purposes:

It is used to create and manage databases. Developers use it to build applications that require data storage.

It performs CRUD operations such as inserting, retrieving, updating, and deleting data.

It manages indexing to improve query performance.

It supports replication and sharding, which are important for high availability and scalability.

It is used in backend development to connect applications (such as Node.js, Python, Java applications) to a database.

It handles real-time data processing and supports large-scale applications.

In short, MongoDB Community Server is used as the backbone of applications that require database functionality.

Uses of MongoDB Compass

MongoDB Compass is mainly used for database visualization and management.

It allows users to create databases and collections easily without writing commands.

It is used to insert documents using a visual editor.

It helps in running queries and applying filters using a graphical interface.

It provides schema analysis, which shows how data is structured inside collections.

It allows users to update and delete documents easily by clicking options.

It is very useful for beginners who are learning MongoDB and do not want to rely only on command-line operations.

Developers also use Compass for debugging and analyzing data during development.

In simple terms, MongoDB Compass is used to simplify database management and make MongoDB easier to use.

Installing Steps for MongoDB

9.1 Installation of MongoDB Community Server (Windows)

Step 1: Download MongoDB Community Server

Visit the official MongoDB website.

Go to the “Download” section and select:

- Version: Latest Stable Version
- Platform: Windows
- Package: MSI

Click on the download button.

Step 2: Run the Installer

After downloading, double-click the .msi file to start installation.

- Click “Next”
 - Accept the License Agreement
 - Choose “Complete” setup
-

Step 3: Install MongoDB as a Service

During installation, select:

- “Install MongoDB as a Service”
- Choose “Run service as Network Service user”

This ensures MongoDB runs automatically in the background.

Click “Next” and then “Install”.

Step 4: Complete Installation

After installation is complete, click “Finish”.

MongoDB will now be installed and running in the background.

If MongoDB shell opens successfully, installation is correct.

9.2 Installation of MongoDB Compass

MongoDB Compass may be included during MongoDB installation. If not, follow these steps:

Step 1: Download MongoDB Compass

Go to the official MongoDB website.

Download MongoDB Compass for your operating system.

Step 2: Install MongoDB Compass

- Open the downloaded file
- Follow installation steps
- Click “Next” and then “Install”

After installation, launch MongoDB Compass.

Images for MongoDB Community :

Or, to install the Community version locally:

Version

8.2.5 (current)

▼

Platform

Windows x64

▼

Package

msi

▼

Download

Copy link

More Options

Images for MongoDB Compass:

Learn more

Version

1.49.1 (Stable)

▼

Platform

Windows 64-bit (10+)

▼

Package

exe

▼

Download

Copy link

More Options

CRUD Operations:

CREATE Operation (Insert Data)

The Create operation is used to insert a new document into a collection.

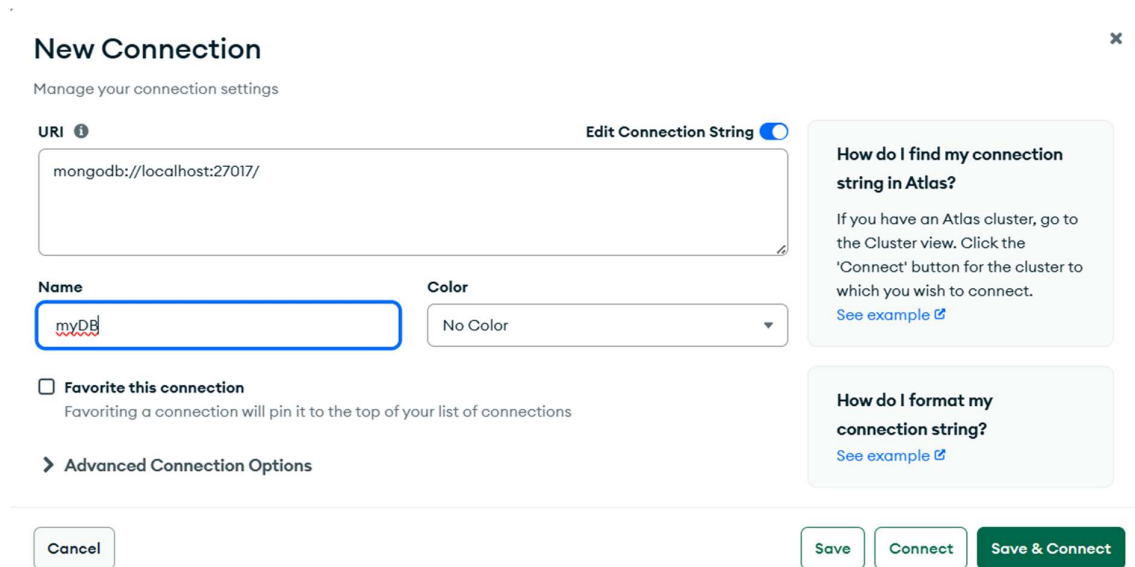
Step-by-Step Procedure (Using Compass)

Step 1: Open MongoDB Compass

Step 2: Connect using:

`mongodb://localhost:27017`

and then click on save and connect Button

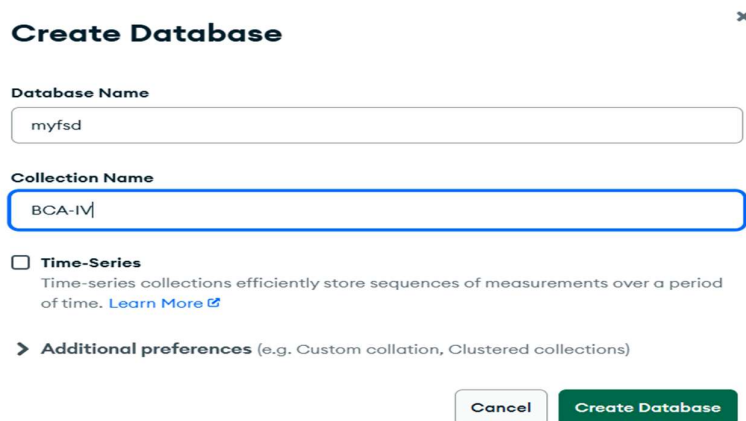


The screenshot shows the 'New Connection' dialog in MongoDB Compass. At the top, it says 'New Connection' with a close button (X) and a subtitle 'Manage your connection settings'. Below this, there's a 'URI' field with a help icon and an 'Edit Connection String' toggle. The URI field contains 'mongodb://localhost:27017/'. Below the URI field, there are two input fields: 'Name' (containing 'myDB') and 'Color' (a dropdown menu showing 'No Color'). There's a checkbox for 'Favorite this connection' with a subtext 'Favoriting a connection will pin it to the top of your list of connections'. Below that is a link for 'Advanced Connection Options'. On the right side, there are two informational boxes: 'How do I find my connection string in Atlas?' and 'How do I format my connection string?'. At the bottom, there are three buttons: 'Cancel', 'Save', and 'Save & Connect'.

Step 3: Click on **Create Database**

Step 4: Enter:

- Database Name: myfsd
 - Collection Name: BCA-IV
- Step 5: Click **Create Database**



The screenshot shows the 'Create Database' dialog in MongoDB Compass. At the top, it says 'Create Database' with a close button (X). Below this, there are two input fields: 'Database Name' (containing 'myfsd') and 'Collection Name' (containing 'BCA-IV'). There's a checkbox for 'Time-Series' with a subtext 'Time-series collections efficiently store sequences of measurements over a period of time. Learn More'. Below that is a link for 'Additional preferences (e.g. Custom collation, Clustered collections)'. At the bottom, there are two buttons: 'Cancel' and 'Create Database'.

Now to insert data:

Step 6: Open students collection

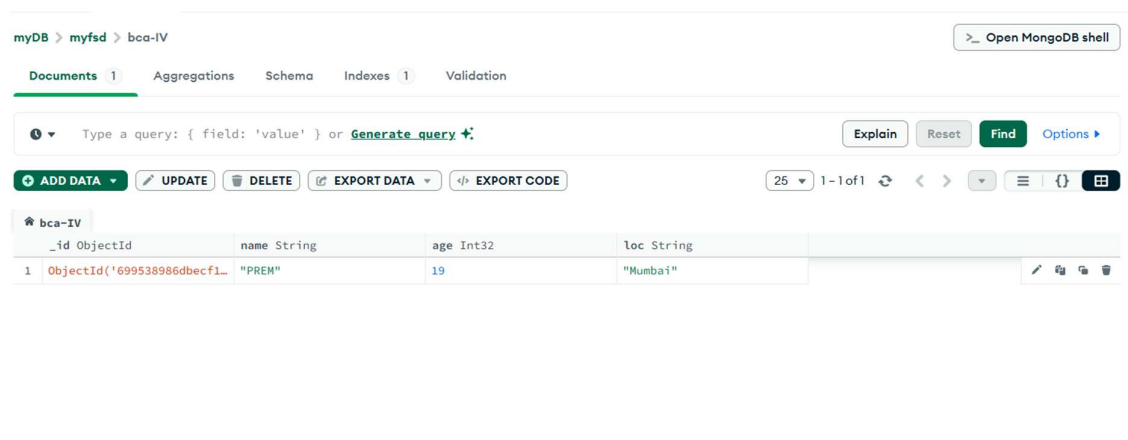
Step 7: Click **Insert Document**

Step 8: Enter the following JSON:

```
{  
  "name": "Prem",  
  "age": 19,  
  "city": "Mumbai"  
}
```

Step 9: Click **Insert**

The document will be saved successfully.



Conclusion:

We learned about MongoDB Community Server, which acts as the core database engine responsible for storing and managing data, and MongoDB Compass, which provides a graphical user interface to interact with the database easily. Understanding the difference between these two components helped in gaining practical knowledge of how MongoDB works internally and externally.

The main objective of this project was to understand and perform CRUD operations — Create, Read, Update, and Delete. Through practical implementation using MongoDB Compass, we successfully created a database and collection, inserted documents, retrieved stored data, updated existing records, and deleted unwanted documents. These operations form the foundation of database management and are essential for building real-world applications.

This project enhanced our understanding of NoSQL databases and demonstrated how MongoDB provides flexibility, scalability, and ease of use compared to traditional relational systems. Overall, MongoDB proves to be a powerful database solution for modern application development.

